

WASP-52b

R-1837

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_160922 · created 2026-08-03T06:33:09+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457653.704 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.064
Transit depth (Rp/Rs)^2	4.66 +/- 2.75 [%]
Orbital Inclination (inc)	86.97 +/- 2.73
Airmass coefficient 1 (a1)	1.23 +/- 0.6
Airmass coefficient 2 (a2)	-0.12 +/- 0.29
Scatter in the residuals of the lightcurve fit is	50.78 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.082 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.064 vs 0.1646 (deviation 0.8σ)
Duration fit vs published	0.082 d vs 0.0758 d (deviation 0.41σ)
Mid-time O–C	7.0 min
Depth SNR	3.4
Residual scatter	50.78 %

Light curve

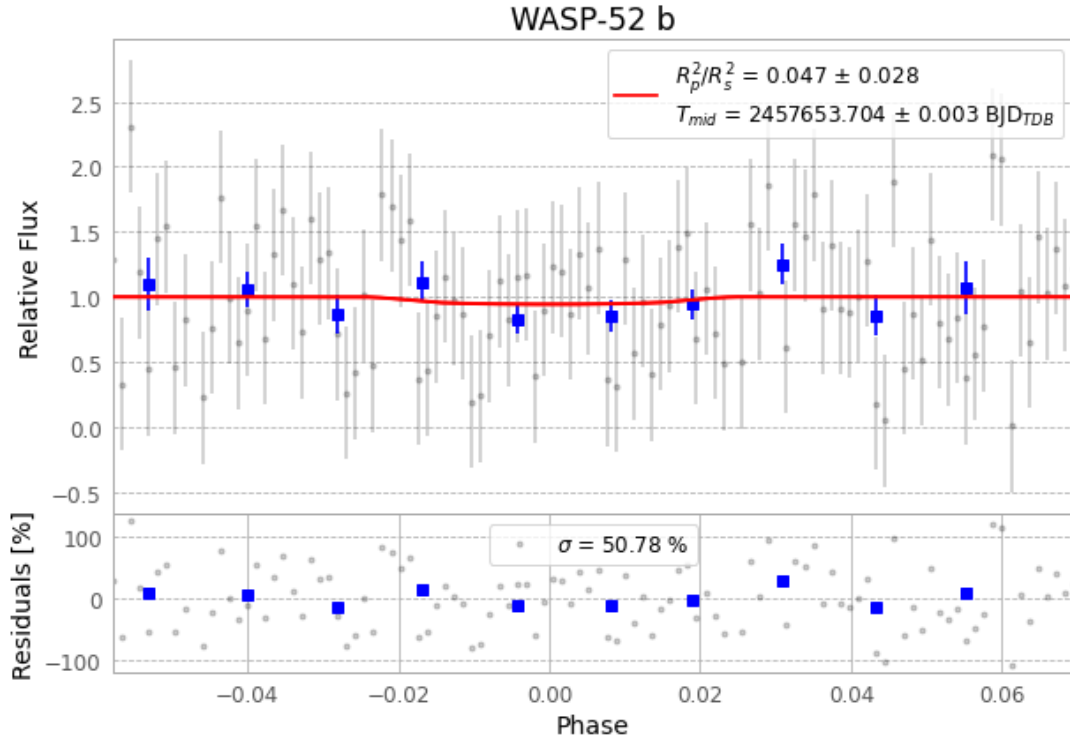


Figure 1 — FinalLightCurve_WASP-52 b_2016-09-21.png (SHA-256 f50507f3d607058a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 69], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f1e381f8a81e92be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

108 FITS frames (71 MB), WASP-52160922021812.FITS → WASP-52160922073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-160922.log (f43a97785455...), FinalLightCurve_WASP-52 b_2016-09-21.png (f50507f3d607...), FinalLightCurve_WASP-52 b_2016-09-21.csv (114c82ced699...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 06:33Z.